SwarmBeauty Theme Demo

A showcase of the all-in-one LaTeX Beautiful Theme

LaTeX Helper Swarm Project

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1 Introduction

This document demonstrates the capabilities of the `swarmbeauty` theme, a comprehensive LaTeX document style designed for clarity, elegance, and usability. The theme is built on top of KOMA-Script and provides a rich set of features out of the box, including beautiful section headings, styled tables, syntax-highlighted code blocks, and custom block environments for notes, tips, and warnings.

The design philosophy behind this theme prioritizes **readability** and **professional aesthetics** while maintaining compatibility with the standard LaTeX ecosystem. Every element—from the title page to the footnote—has been carefully crafted to produce documents that look polished and modern.

1.1 Motivation

Writing beautiful LaTeX documents typically requires dozens of package inclusions, meticulous configuration of colors, fonts, and spacing, and a deep understanding of the underlying typesetting system. The `swarmbeauty` theme aims to eliminate this friction by providing sensible defaults that work well for academic papers, technical reports, and documentation.

> **Note**
>
> This theme requires **LuaLaTeX** or **XeLaTeX** with the `-shell-escape` flag for the minted code highlighting.

2 Typography and Text

The theme uses **Latin Modern** fonts throughout, with *italic*, **bold**, and ***bold italic*** variants. Mathematics is rendered using `unicode-math` for consistent glyph quality. Microtypographic improvements via `microtype` provide better character spacing and hyphenation.

2.1 Inline Elements

You can highlight **important terms** using the `hltext` command. For inline code, use the `code` command like this: `some_function()`. For code with underscores, use `text_with_underscores`. Emphasis boxes are available too: **key concept**.

2.2 Lists

- First item in an unordered list
- Second item with **emphasis**
- Third item with nested content:
  - Nested item A
  - Nested item B

1. Step one: read the documentation
2. Step two: include the theme
3. Step three: write beautiful documents

3 Block Environments

The theme provides five distinct block environments, each with a unique color accent and left border for quick visual identification.

Note

This is a **note** block. Use it for general observations, additional context, or supplementary information that is relevant but not critical to the main argument.

Tip

This is a **tip** block. Use it to highlight best practices, shortcuts, or recommendations that can improve the reader's workflow or understanding.

Warning

This is a **warning** block. Use it to alert the reader about potential pitfalls, common mistakes, or deprecated features that they should be aware of.

Important

This is a **danger** block. Use it for critical warnings about data loss, security vulnerabilities, or operations that cannot be undone.

Example

This is an **example** block. Use it to present worked examples, sample output, or illustrations of the concepts being discussed.

4 Tables

The theme includes custom table rules and integrates seamlessly with both `booktabs` and `tabularray`. Below are examples of each style.

4.1 Booktabs Table

Table 1 Comparison of LaTeX compilation engines

Engine	Speed	Font Support	Unicode
pdfLaTeX	Fast	Limited (Type 1)	Partial
XeLaTeX	Medium	Full (OpenType)	Full
LuaLaTeX	Slow	Full (OpenType)	Full

4.2 Tabularray Table

Table 2 Key packages used by the theme

Package	Purpose	Required
fontspec	OpenType font loading	Yes
microtype	Microtypographic improvements	Yes
minted	Pygments-based code highlighting	Yes
tcolorbox	Styled boxes and code blocks	Yes
booktabs	Professional table rules	Yes
tabularray	Modern table formatting	Optional
tikz	Graphics and decorations	Yes

5 Code Listings

The `codeblock` environment provides syntax-highlighted code with line numbers, powered by the Pygments library through the `minted` package.

Fibonacci sequence generator

```
def fibonacci(n: int) -> list[int]:
    """Generate the first n Fibonacci numbers."""
    seq = [0, 1]
    for i in range(2, n):
        seq.append(seq[-1] + seq[-2])
    return seq[:n]

if __name__ == "__main__":
    result = fibonacci(10)
    print(f"First 10: {result}")
```

A simple equation

```
\begin{equation}
  E = mc^2
  \label{eq:einstein}
\end{equation}
```

6 Mathematics

The theme provides styled theorem environments with colored left borders.

Theorem 6.1: Euler's Identity

For any real number x, the following identity holds:

$$e^{i\pi} + 1 = 0$$

Definition 6.1: Metric Space

A *metric space* is a pair (X, d) where X is a set and $d : X \times X \to \mathbb{R}_{\geq 0}$ satisfies:

1. $d(x, y) = 0 \iff x = y$ (identity of indiscernibles)
2. $d(x, y) = d(y, x)$ (symmetry)
3. $d(x, z) \leq d(x, y) + d(y, z)$ (triangle inequality)

Lemma 6.1: Nested Intervals

Let $\{[a_n, b_n]\}_{n=1}^{\infty}$ be a sequence of closed, bounded intervals such that $[a_{n+1}, b_{n+1}] \subseteq [a_n, b_n]$ for all n. Then $\bigcap_{n=1}^{\infty}[a_n, b_n] \neq \emptyset$.

6.1 Inline and Display Math

The quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

A more complex display equation:

$$\int_{-\infty}^{\infty} e^{-x^2}\, dx = \sqrt{\pi}$$

7 Figures

Figures use standard `figure` environments with the theme's caption styling.

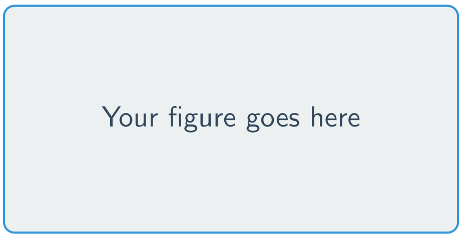

Figure 1 A placeholder figure rendered with TikZ

Tip

Replace the placeholder above with `includegraphics[width=linewidth]{image.pdf}`.

8 Summary

This demo has showcased the following features of the `swarmbeauty` theme:

1. **Title page** with colored header bar and metadata box
2. **Section headings** with decorative rules
3. **Block environments** (note, tip, warning, danger, example)
4. **Styled tables** with booktabs and tabularray
5. **Syntax-highlighted code** via minted + tcolorbox
6. **Theorem environments** with colored borders
7. **Typography** with Latin Modern fonts and microtype
8. **Headers and footers** with page numbers and document info

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Generated with `swarmbeauty` theme — LaTeX Helper Swarm Project